SQL

Ask for data, not the procedure to find it

THE MENTAL MODEL

Tell the librarian which books you want; they choose the search path.

QUESTION → QUERY → PLANNER → ROWS

READ THE CODE

```
SELECT department, COUNT(*) AS n
FROM employees
WHERE active = TRUE
GROUP BY department
ORDER BY n DESC;
```

FOUR CLUES TO READ IT

- SELECT names columns to return.
- FROM identifies the table.
- WHERE filters rows before grouping.
- GROUP BY / ORDER BY aggregate and sort results.

WHAT MAKES IT DIFFERENT

Structured Query Language (SQL) is declarative: describe the result.

Database engines choose execution plans and use indexes when useful.

WHERE IT FITS

Querying relational databases, reporting, data transformations.

WATCH OUT

The example assumes an employees table; order is not guaranteed without ORDER BY.

TRY ALONE • 60 SEC

Change DESC to ASC. Which departments appear first?